

--	--	--	--	--	--	--	--

B. Tech. Degree IV Semester Examination April 2016

CS 1406 DATA COMMUNICATIONS

(2012 Scheme)

Time: 3 Hours

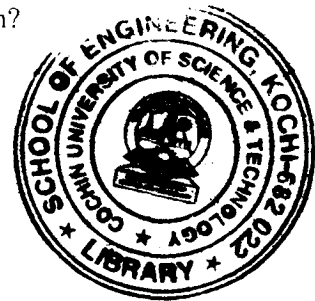
Maximum Marks: 100

PART A

(Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) What are the components of data communication?
 (b) What are the advantages of digital transmission over analog transmission?
 (c) What is delta modulation? Write its advantage over PCM.
 (d) Write about different scrambling techniques.
 (e) Explain the frame structure of HDLC.
 (f) Write about selective repeat ARQ protocol.
 (g) Explain TDM. Why synchronization is important in TDM?
 (h) What is DOCSIS?



PART B

(4 × 15 = 60)

- II. (a) What is transmission medium? What are the different types of transmission medium? (10)
 (b) Explain briefly composite signals. (5)

OR

- III. (a) Describe the different types of transmission impairments. (8)
 (b) What are the theoretical formulas developed to calculate data rate for a noiseless and noisy channel? (7)

- IV. (a) Explain different line coding schemes by using the data 11001110. (8)
 (b) What is the difference between FSK and FM? Explain it with the help of band width requirement. (7)

OR

- V. (a) Explain LZW coding. (10)
 (b) What is constellation diagram? Draw the constellation diagram of ASK, PSK and QAM. (5)

- VI. (a) What is frequency dependent coding? Explain it with an example. (10)
 (b) Compare synchronous and asynchronous transmission techniques. (5)

OR

- VII. (a) How is single bit error detecting and correcting in data communication? (8)
 (b) Explain the following. (7)
 (i) stop-and-Wait ARQ
 (ii) GO-Back-N ARQ

- VIII. (a) What are the differences between IEEE 802.11 FHSS and IEEE 802.11 DHSS? (8)
 (b) Why is multiplexing needed in data communication? (7)

OR

- IX. (a) What are the major components of telephone network? (8)
 (b) Distinguish between traditional cable network and HFC network. (7)